Identification of liquid organic compounds

* carboxylic acid → (COOH) group

aliphatic → aliphatic organic compounds

- monobasic acid
  - HCOOH formic acid
  - CH_3COOH acetic acid
- dibasic acid
  - oxalic acid
  - COOH × COOH
  - CH_2-COOH × CH_2-COOH
  - Succinic acid
- tribasic acid
  - CH_2-COOH, HO-C-COOH, CH_2-COOH
  - citric acid
  - HO-CH-COOH, HO-CH-COOH
  - tartaric acid

aromatic

* Physical Properities
  1. colour
  2. state
  3. miscibility
  4. Effect on litmus paper:-

Chemical Properities:-

acidity test:- $2R\text{-}COOH + Na_2CO_3 \rightarrow 2R\text{-}COONa + H_2CO_3 \rightarrow H_2O + CO_2$ → Evolution and effervesence of CO_2

- → $\frac{1}{2}$ gm of Na_2CO_3
- → 1ml of unknown liquid

oxidation test:-

- → 1ml of acid
- → 1ml of Na_2C
- → 1ml of $KMnO_4$

→ change in colour of $KMnO_4$ (HCOOH) formic acid

→ no change of colour → CH_3COOH acetic acid